

used to make a Windows, Linux or Mac sketch as well. The main difference in interaction is that there is no concept of a hover or 'mouse over' in a touchscreen Android device. Keep that in mind while writing sketches—mouseX and mouseY in the code above are the coordinates where the touch event occurred.

In sequence, the code can be broken down as shown below:

1. void setup() This section runs only once—at the beginning of the sketch.
 - a. Size (width and height): size of the sketch window.
 - b. Background (colour value): the background colour—if it's a single value, Processing considers it a shade of grey. The value can also be an R,G,B value of (255,255,255), which would give the background a white colour.
 - c. Smooth(): This allows the smooth rendering of the window.
 - d. noStroke(): This will ensure the shapes have no border.
2. void draw() As mentioned earlier, this is an infinite loop.
 - a. mousePressed: This is a Processing constant that is attached to the mouse press (a.k.a Touch) event. This line is the gatekeeper to the rest of our code—it runs our logic only if the user has touched the screen.
 - b. Fill (r value, g value, b value): Fill() will set the brush colour in RGB mode.
 - c. Random (value): This is a function that picks a random value from the range specified. For example, when we specify only a single number, it will consider the range to be from zero to that number. Random(255) will select a number at random every time from 0 - 255. By combining fill() and random(), we can achieve random colours to make our sketch look more interesting.
 - d. Ellipse (center X, center Y, width, height): The ellipse() function draws an ellipse with its centre being the values x and y, with the width and height as we specify them. Here, too, we have used random() to make the sketch livelier.

Running the sketch

Click the button that looks like Play at the top to run the sketch. Processing will automatically create an Android emulator and run our sketch in it. In case it creates and starts the emulator but gives an error while running, I suggest the following: leave the emulator open, and re-run the sketch. Alternatively, you can open the Android AVD Manager from the SDK Manager (Tools > Manage AVDs...) and start the emulator first before running your sketches. Once a sketch is running, you can exit it by using the 'Back' button in the emulator or device. You can leave

the emulator running as you code and run more sketches—this is to avoid starting the emulator and booting it every time you want to run a sketch.

Signing and packaging your sketch

So, we have made a sketch and it's time to share it with the rest of the world. Signing and packing your sketch for the market, now known as Google Play, is a bit of a lengthy procedure. I will show you how to do it below. Before we begin, let us export our sketch in Processing. From within the IDE, select *File > Export Android Project*. This will generate the source files we require to prepare our package. Change to the export directory (Processing opens that directory for us when we run the export command). All our commands will run from there.

1. Generating a public-private key pair—We will do this by executing the following at the terminal:

```
keytool -genkeypair -v -keystore my-release-key.keystore -alias mykey1 -keyalg RSA -keysize 2048 -validity 10000
```

Keytool is a utility that is part of the JDK. The above command generates an RSA encrypted public-private key pair valid for 10,000 days and it is stored in a keystore file called 'my-release-key.keystore'. Upon running the above command, the following output is generated, asking us certain questions:

```
Enter keystore password:
Re-enter new password:
What is your first and last name?
  [Unknown]: Siddharth
What is the name of your organizational unit?
  [Unknown]: Personal
What is the name of your organization?
  [Unknown]: Personal
What is the name of your City or Locality?
  [Unknown]: Pune
What is the name of your State or Province?
  [Unknown]: Maharashtra
What is the two-letter country code for this unit?
  [Unknown]: IN
Is CN=Siddharth, OU=Personal, O=Personal, L=Pune, ST=Maharashtra,
C=IN correct?
[no]: yes
```

```
Generating 2,048 bit RSA key pair and self-signed certificate
(SHA1withRSA) with
a validity of 10,000 days
for: CN=Siddharth, OU=Personal, O=Personal, L=Pune,
ST=Maharashtra, C=IN
```

```
Enter key password for <mykey1>
(RETURN if same as keystore password):
[Storing my-release-key.keystore]
```